Big Data Analytics

CISELEC 501A-CS41S1

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Laboratory Activity #04: CRISP-DM | Modeling with WEKA

Total Points: 210

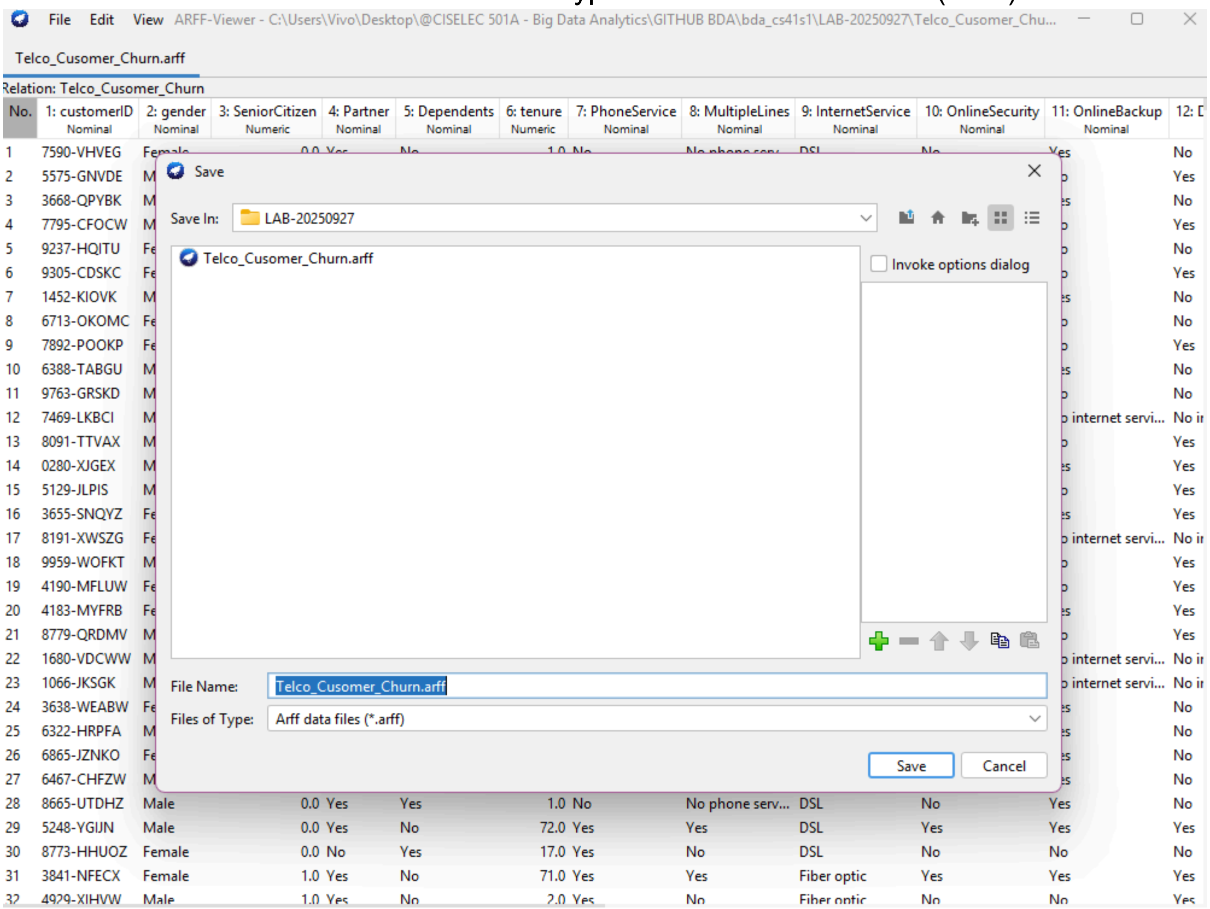
OBJECTIVE: This laboratory exercise aims to help students understand how proper data preparation supports effective data modeling. By converting the dataset, selecting key attributes, and applying various classification algorithms, students will observe how data quality and preparation choices directly affect model performance. This activity emphasizes the critical role of the data preparation phase in producing reliable, accurate, and meaningful modeling results.

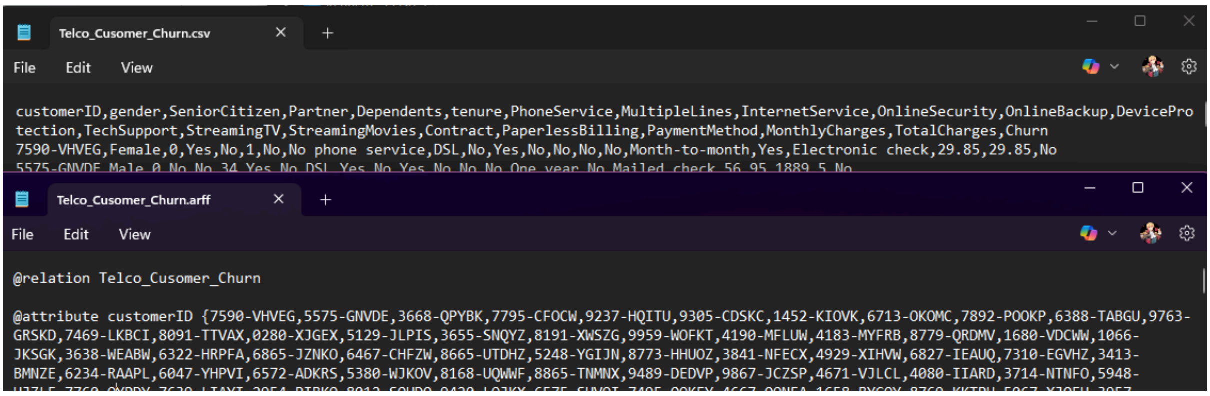
I. DATA UNDERSTANDING

A. Download the “Telco_Cusomer_Churn.csv” file from our GIT repository inside “20250927-Lab-Wk10” folder.

B. Convert the downloaded .csv file to .arff (5 pts)

1. Open WEKA, go to the “Tools” menu and click “ArffViewer”.
2. The “ARFF-Viewer” window will open. Click File >> Open. Note: Ensure that “Files of Type” is set to “CSV data files (*.csv)”.
3. Locate and open the file
4. Once the file is loaded in the “ARFF-Viewer” window, click File >> Save As. Note: Ensure that “Files of Type” is set to “Arff data files (*.arff)”.





C. REFLECTION

1. Compared to the original file format (.csv), what changes were made after converting the file to .arff? Kindly elaborate on your observation. (**10 pts**)
    a) In terms of viewing it in a graphical user interface, there are no changes whatsoever. Datas are not tampered nor the alignment and other aspects of information it has.
    b) In terms of binaries, there is however changes but it does not affect the data of the dataset.
    c) As you can see on the image above, the contents differ from one another. CSV files as we know uses ‘,’ char to split it so its still a typical format.

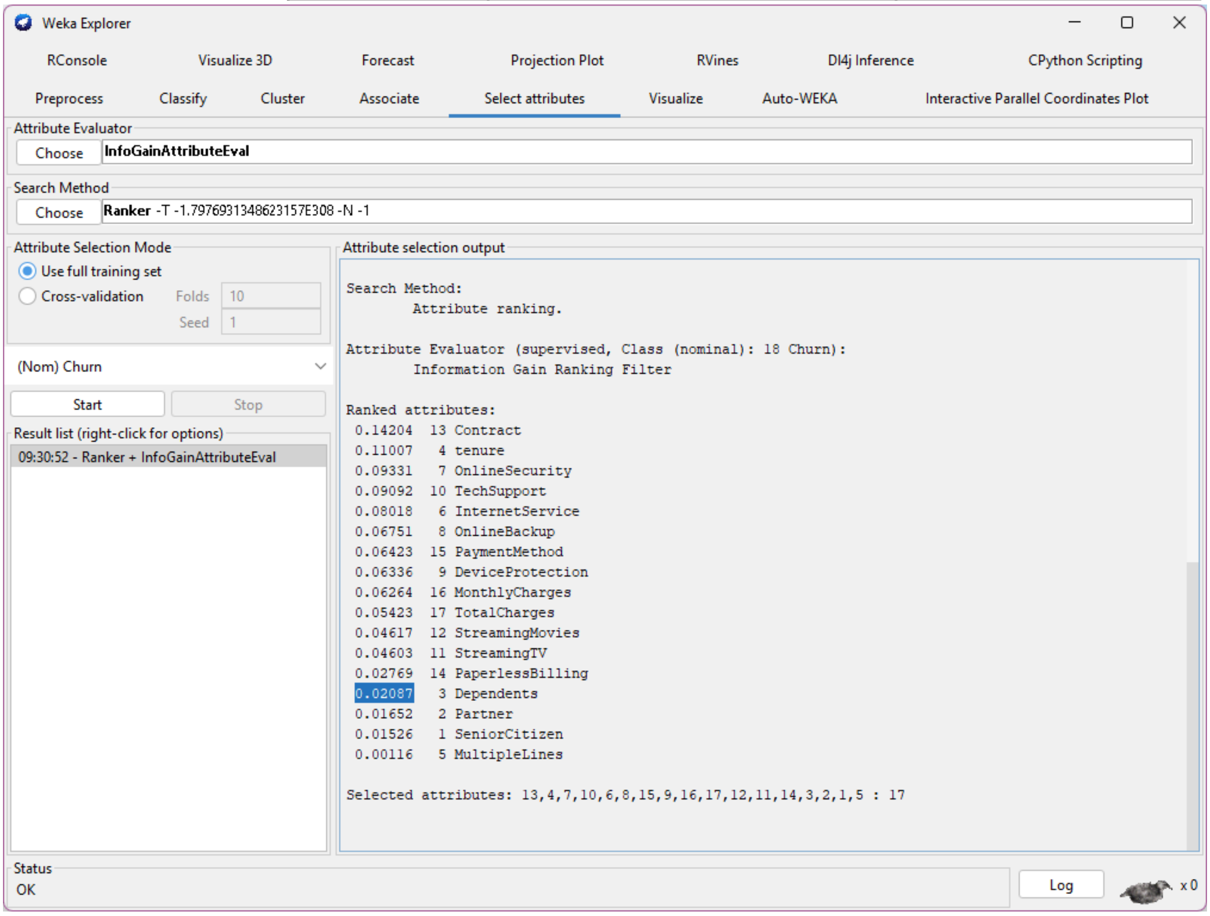
       However, on our converted .arff, we see tags and symbols such as @relation and @attribute. This could prove helpful specially for searching attributes etc…

II. DATA PREPARATION

A. Click the “Explorer” button in the WEKA main window.
B. Under the “Process” tab, load the newly created .arff file.
C. Once the file is loaded, proceed to the “Select attributes” tab and configure the following:
    1. Attribute Evaluator: InfoGainAttributeEval
    2. Search Method: Ranker
    3. Attribute Selection Mode: Use full training set
    4. Class name: (Nom) Churn
D. Press the “Start” button to view the ranking of all the attributes.
E. Under the “Attribute selection output” section, locate the “Ranked attributes” label and fill in the table below (identify the top 14 attributes, from highest to lowest). (10 pts)

Rank No.	Attribute Name	Score
1	Contract	0.14204
2	tenure	0.11007
3	OnlineSecurity	0.09331
4	TechSupport	0.09092
5	InternetService	0.08018
6	OnlineBackup	0.06751
7	PaymentMethod	0.06423
8	DeviceProtection	0.06336
9	MonthlyCharges	0.06264

10	TotalCharges	0.05423
11	StreamingMovies	0.04617
12	StreamingTV	0.04603
13	PaperlessBilling	0.02769
14	Dependents	0.02087



F. REFLECTION:

1. What could be the impact if different field were selected under the "Class name"? (**5 pts**)
   a) There is a lot of impact. What we learned in the previous discussion is that this evaluator is one of the algorithms that helps us identify features, to gain information about features. Just like the discussed algorithm "chi squared test". This classifies our features. Therefore we say "what feature is relevant with our target"
   b) This will simply mean that if we change Class Name, we are changing our target. Instead of finding features that are relevant with our Churn, it will instead find out what features relevant to the changed target. (One I'm interested in is the Contract attribute hehehe.)
2. Explain why it is important to select the field containing the labeled data during the Feature Selection stage. (**15 pts**)
   a) As per the previous question, it is important as it will be the basis of what we should target.
   b) Without selecting it, then those attributes are relevant to what? This could also answer the question of one of your objectives under the Business Understanding Phase.

   Therefore, Choosing a correct label data ensures that selected features are really meaningful for our actual problem. This will avoid waste effort and actually avoid being misled on insights and performance.

III. MODELING

A. While the .arff file is loaded, proceed to the "Clasify" tab and configure the following:
   1. Classifier: classifiers >> trees >> J48
   2. Test options: Use training set
   3. More options: Output entropy evaluation measures (Enable)
   4. Class: (Nom) Churn

B. Perform the classification by pressing the "Start" button.

C. Locate the "Summary" label inside the "Classifier output" section and complete the table below: (10 pts)
   1.

Correctly Classified Instances	6047 85.8583 %
Incorrectly Classified Instances	996 14.1417 %
Kappa statistics	0.6101
Precision (Class: NO)	0.874
Precision (Class: YES)	0.799
Recall (Class: NO)	0.943
Recall (Class: YES)	0.624
F-Measures (Class: NO)	0.907
F-Measures (Class: YES)	0.701

D. Repeat steps 1 to 3 using the following Classification algorithms and provide the same table as in step #3:
   1. classifiers >> bayes >> NaiveBayes
   2. classifiers >> rules >> DecisionTable
   3. classifiers >> functions >> LibSVM

E. Repeat steps 1 to 4, but this time load the top 7, 10, and 14 attributes identified in II. DATA PREPARATION, item #5. Don't forget to create the same table in step #3 (under III. MODELING) NOTE:
   1. Removal of some dataset fields can be done inside the "Preprocess" tab, under the "Attributes section". Check the selected fields then press the "Remove" button. If you wish to proceed with another set of fields, just reload the same file inside the "Preprocess" tab.
   2. Ensure that you DO NOT REMOVE the "Churn" attribute, and make sure it is included during model evaluation.
   3. Kindly use the table presented in ANNEX-01.

A. ANNEX-01: Model Evaluation (145 pts)

CLASSIFIERS	Trees >> J48				Bayes >> NaiveBaiyes				Rules >> DecisionTable				Functions >> LibSVM			
ATTRIBUTES	All	Top 14	Top 10	Top 7	All	Top 14	Top 10	Top 7	All	Top 14	Top 10	Top 7	All	Top 14	Top 10	Top 7
Correctly Classified Instances	6047	5965	5821	5687	5125	5122	5197	5277	5677	5677	5632	5643	6375	6412	6465	5613
Incorrectly Classified Instances	996	1078	1222	1356	1918	1921	1846	1766	1366	1366	1411	1400	668	631	578	1430
Kappa statistics	0.6101	0.5825	0.5277	0.4754	0.4189	0.4186	0.4287	0.4425	0.4678	0.4678	0.4349	0.4402	0.7402	0.7556	0.7779	0.4291
Precision (Class: NO)	0.874	0.871	0.858	0.846	0.908	0.908	0.902	0.899	0.843	0.843	0.830	0.831	0.906	0.911	0.919	0.829
Precision (Class: YES)	0.799	0.760	0.711	0.668	0.492	0.491	0.504	0.519	0.669	0.669	0.671	0.674	0.901	0.908	0.913	0.662
Recall (Class: NO)	0.943	0.930	0.915	0.902	0.700	0.699	0.722	0.743	0.905	0.905	0.915	0.915	0.971	0.973	0.974	0.911
Recall (Class: YES)	0.624	0.618	0.583	0.545	0.803	0.804	0.783	0.768	0.532	0.532	0.480	0.486	0.722	0.737	0.764	0.480
F-Measures (Class: NO)	0.907	0.899	0.886	0.873	0.791	0.790	0.802	0.813	0.873	0.873	0.870	0.871	0.938	0.941	0.946	0.868
F-Measures (Class: YES)	0.701	0.682	0.641	0.600	0.610	0.610	0.613	0.619	0.593	0.593	0.560	0.565	0.802	0.814	0.832	0.556

B. REFLECTION:

a. Based on the assessment provided in ANNEX-01, is there any relevance in performing Feature selection prior to the Modeling phase? Elaborate your answer. (10 pts)

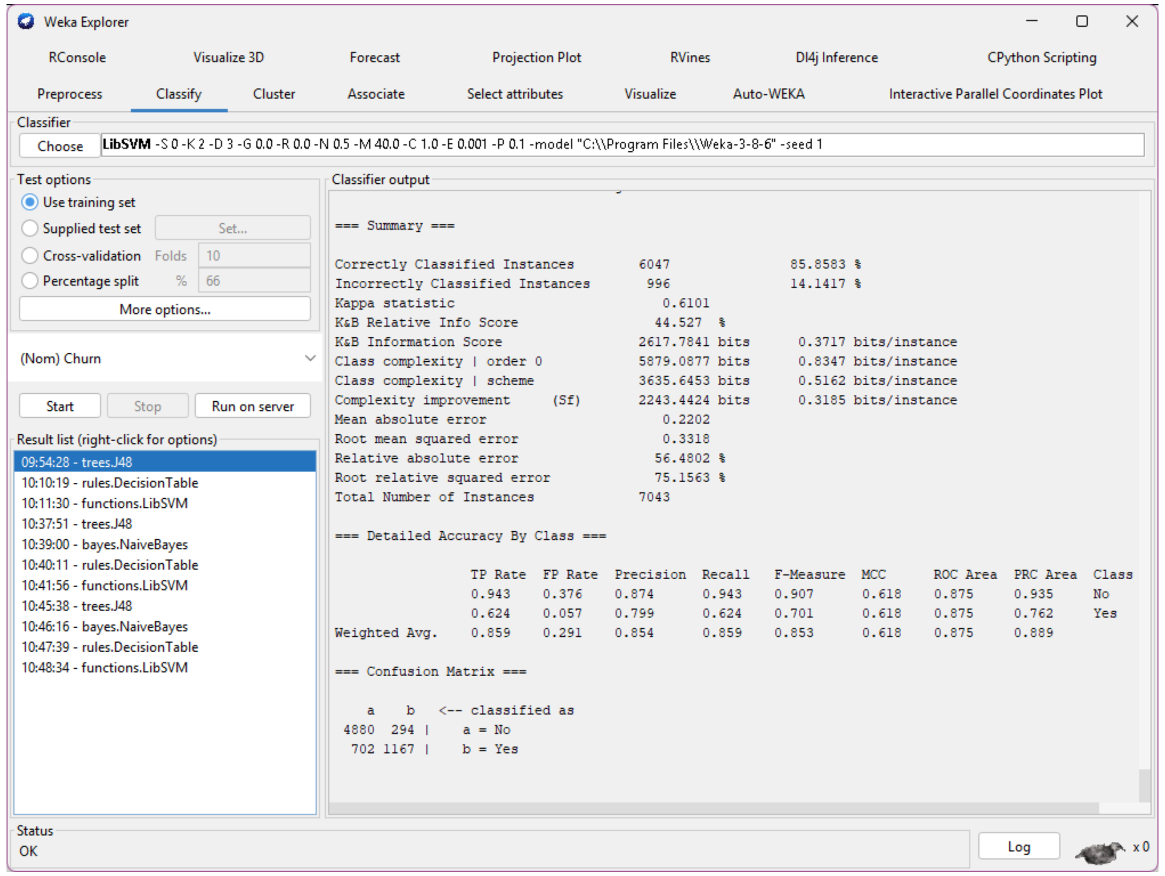
i. There is a lot of relevance. It is so highly important. We can clearly see on the ANNEX-01 Table that the algorithm varies and selects features. There are a lot of major changes every time you drop or just select the features that you want. It really highlights different measurements such as Classified Instances, Recall and F measures.

ii. Look at LIBSVM. When Selecting the top 10 Features. It really outperforms everyone, but when selecting the top 7 features from the InfoGain evaluator, it drastically changes. This really tells us that we can use other evaluations during our feature selection phase. How about using chi squared, or even reliefF? and then decide amongst those which give meaningful information. It may even outperform the current attribute selection we have chosen.

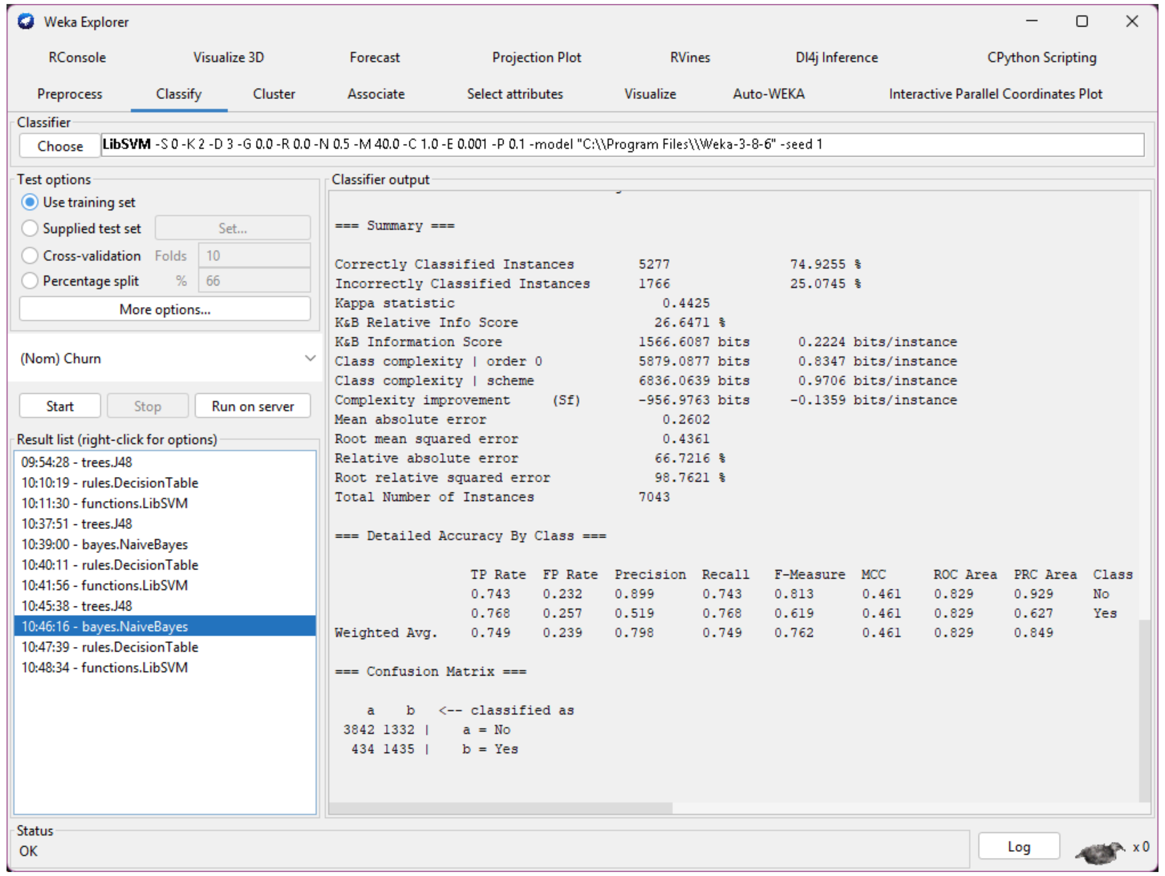
iii. But in this instance, we used the top 10 attributes from InfoGainAttributeEval and fed it into LibSVM that outperforms the other 3 algorithms.

A. IMAGES OF THE TOP SCORERS OF EACH ALGORITHM

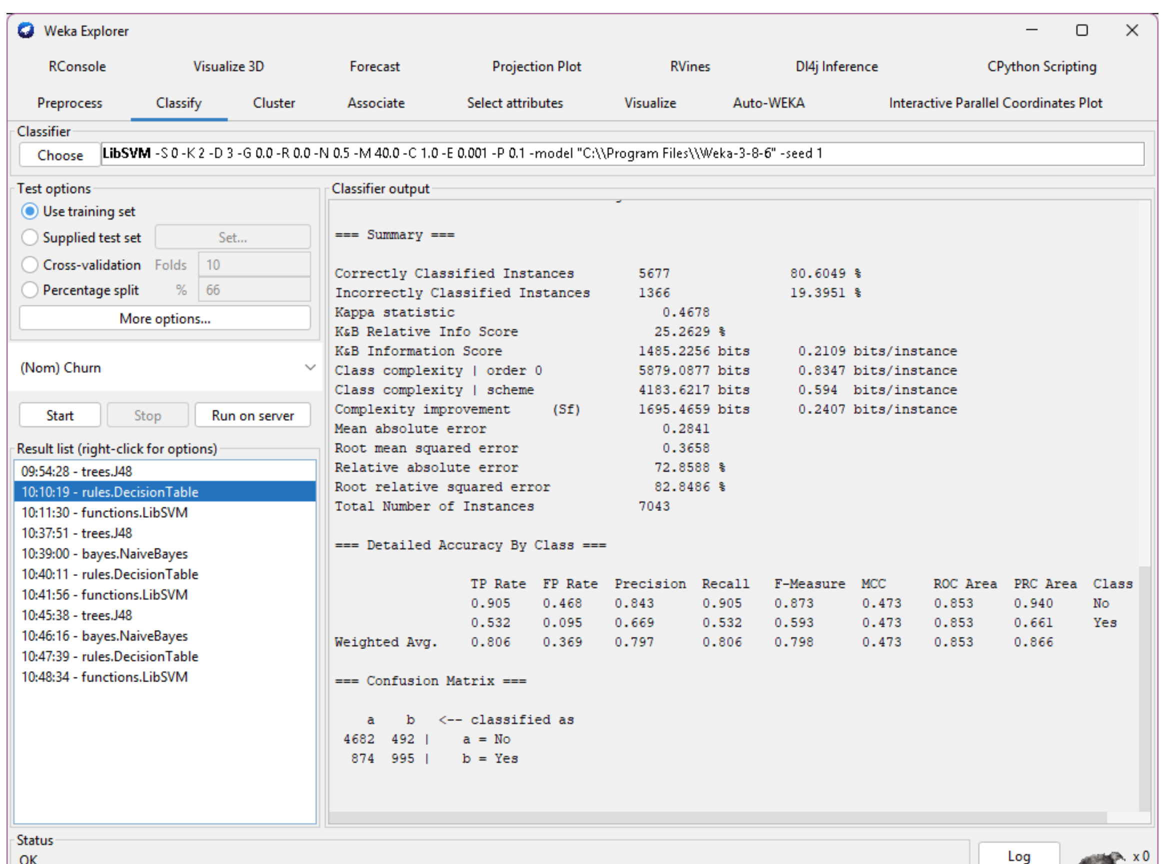
a. J48



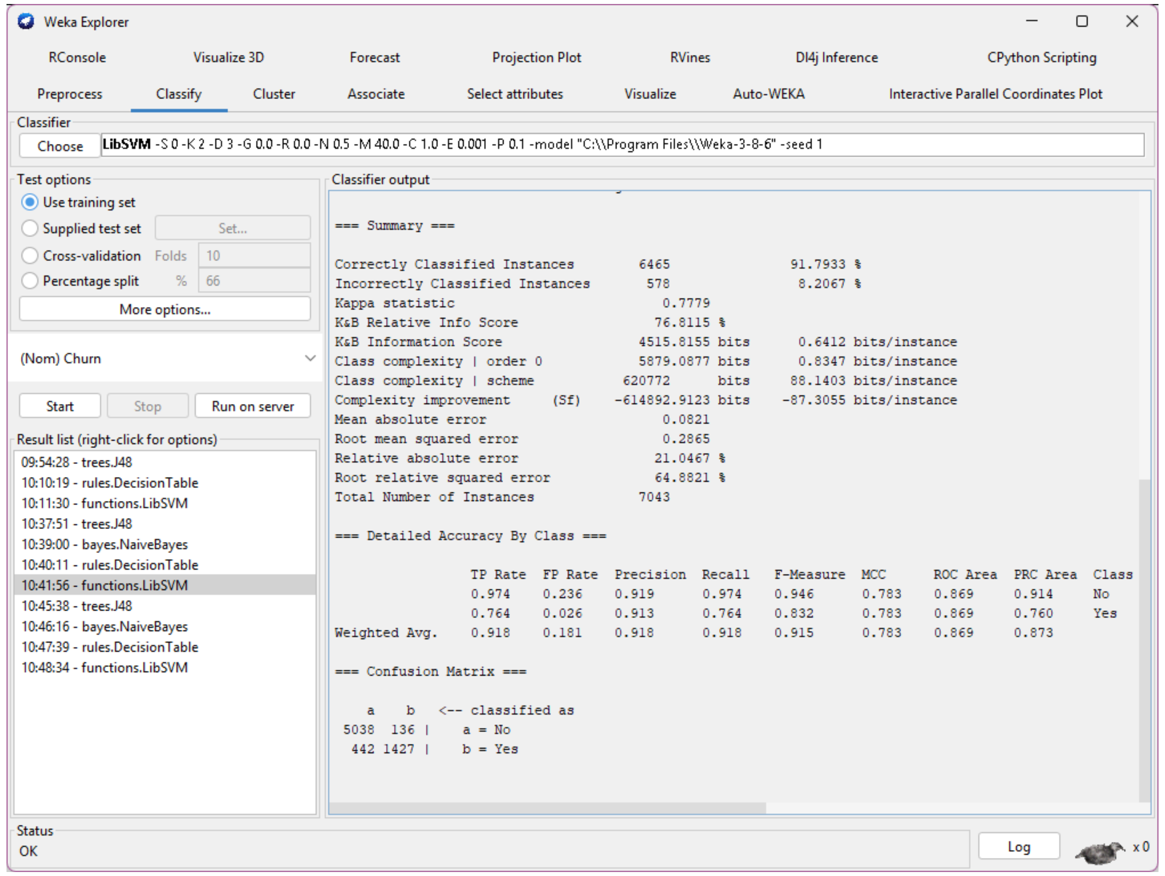
b. NaiveBayes



c. DecisionTable



d. LibSVM


Weka Explorer
RConsole
Visualize 3D
Forecast
Projection Plot
RVines
Dl4j Inference
CPython Scripting
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Auto-WEKA
Interactive Parallel Coordinates Plot
Classifier
Choose
LibSVM -S 0 -K 2 -D 3 -G 0.0 -R 0.0 -N 0.5 -M 40.0 -C 1.0 -E 0.001 -P 0.1 -model "C:\\Program Files\\Weka-3-8-6" -seed 1
Test options
Use training set
Supplied test set
Set...
Cross-validation Folds 10
Percentage split % 66
More options...
(Nom) Churn
Start
Stop
Run on server
Result list (right-click for options)
09:54:28 - trees.J48
10:10:19 - rules.DecisionTable
10:11:30 - functions.LibSVM
10:37:51 - trees.J48
10:39:00 - bayes.NaiveBayes
10:40:11 - rules.DecisionTable
10:41:56 - functions.LibSVM
10:45:38 - trees.J48
10:46:16 - bayes.NaiveBayes
10:47:39 - rules.DecisionTable
10:48:34 - functions.LibSVM
Classifier output
=== Summary ===
Correctly Classified Instances 6465 91.7933 %
Incorrectly Classified Instances 578 8.2067 %
Kappa statistic 0.7779
K&B Relative Info Score 76.8115 %
K&B Information Score 4515.8155 bits 0.6412 bits/instance
Class complexity | order 0 5879.0877 bits 0.8347 bits/instance
Class complexity | scheme 620772 bits 88.1403 bits/instance
Complexity improvement (Sf) -614892.9123 bits -87.3055 bits/instance
Mean absolute error 0.0821
Root mean squared error 0.2865
Relative absolute error 21.0467 %
Root relative squared error 64.8821 %
Total Number of Instances 7043
=== Detailed Accuracy By Class ===
TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class
0.974 0.236 0.919 0.974 0.946 0.783 0.869 0.914 No
0.764 0.026 0.913 0.764 0.832 0.783 0.869 0.760 Yes
Weighted Avg. 0.918 0.181 0.918 0.918 0.915 0.783 0.869 0.873
=== Confusion Matrix ===
a b <-- classified as
5038 136 | a = No
442 1427 | b = Yes
Status
OK
Log
x 0

" I affirm that I have not given or received any unauthorized help in this assignment, and that this work is my own "

Kenneth Arias